

FCC Part 100 Delta & Certification Audit

Part 25 to Part 100 Transition Readiness Assessment (FCC 25-69, SB Docket 25-306)

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System Name: STARLINK-1008 (NORAD #44714) (SpaceX (Space Exploration Technologies Corp.))

FCC Classification: NGSO Satellite System — LEO (386 km)

Constellation Size: 7500 authorized / 2500 deployed / 800.0 kg wet mass

Regime Evaluated: FCC Part 100 (Adopted — FCC 25-69)

4

Certifications Passed

0

Certifications Failed

10

Attestation Needed

1

Reviews Triggered

\$6,296,296

Part 100 Bond Due

Actionable Regulatory Risk Flags:

• 10 certification(s) require additional data or attestation

• 1 Targeted Review Category(ies) triggered

1. Part 25 vs Part 100 Regulatory Delta Matrix

Statutory comparison of key operational and procedural rules affecting this satellite authorization.

Regulatory Dimension	Part 25 (Legacy Framework)	Part 100 (Adopted Framework)
Application Structure	Form 312 + Schedule S	Form 312 Main + Schedule O/F/B

Review Methodology	Bespoke narrative engineering review	Certification-based "Default to Yes"
Public Notice Window	30 Days	15 Days (Single period)
License Duration	15 years	20 years
Surety Bond Requirement	\$5,000,000 (Escalating)	\$6,296,296 (De-escalating / PR only)
De-orbit Timeline	25-Year rule	5-Year post-mission (§ 100.260(e))
Physical Trackability	Statement only	≥10cm LEO / ≥1m >2000km (Affirmative)
Space Safety Reporting	None	Semi-annual mandatory report (NGSO)

2. Surety Bond & Milestone Impact (§ 100.148 & § 100.147)

Part 25 Legacy Bond \$5,000,000 Mandatory escalating bond required for all space station authorizations.	Part 100 Adopted Bond \$6,296,296 Formula: $B = \\$10M - \\$10M \times (D / (0.9 \times A))$ Current Deployment: 33.33% (Fully released at 90%)
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3. Part 100 Certification Readiness Matrix (Schedule O & F)

Deterministic evaluation of all statutory bright-line certifications based on supplied mission parameters.

ID	Statutory Criterion	FCC Section	Status	Evaluation Detail
NGSO-03	Trackability — smallest dimension meets minimum size	§ 100.111(c)(2)(iii)	PASS	Value: 100.0 cm LEO (386 km): 100.0 cm ≥ 10 cm threshold — PASS
NGSO-10	De-orbit within 5 years post-mission	§ 100.260(e)	PASS	Value: 0.8 years Estimated de-orbit: 0.8 years ≤ 5-year threshold — PASS
NGSO-01	Operates only in non-geostationary orbit	§ 100.111(c)(2)(i)	PASS	Value: LEO Verified against telemetry model
NGSO-02	Identifiable by unique signal-based telemetry marker	§ 100.111(c)(2)(ii)	ATTEST	Requires operator attestation — not auto-evaluable
NGSO-04	Will assess/mitigate collision risk on conjunction warning	§ 100.111(c)(2)(iv)	ATTEST	Requires operator attestation — not auto-evaluable
NGSO-05	Small debris collision probability ≤ 0.01	§ 100.111(c)(2)(v)	ATTEST	Requires NASA DAS or equivalent calculation — future ODAR engine
NGSO-06	Large object collision probability ≤ 0.001	§ 100.111(c)(2)(vi)	ATTEST	Requires NASA DAS or equivalent calculation — future ODAR engine
NGSO-07	Human casualty risk from re-entry ≤ 0.0001	§ 100.111(c)(2)(vii)	ATTEST	Requires NASA DAS casualty risk analysis — future ODAR engine
NGSO-08	All stored energy removed at end of life	§ 100.111(c)(2)(viii)	ATTEST	Requires passivation plan review — operator attestation
NGSO-09	Disposed of via atmospheric re-entry	§ 100.111(c)(2)(ix)	PASS	Value: Altitude: 386 km LEO spacecraft — atmospheric re-entry expected
NGSO-11	Probability of successful disposal ≥ 0.9	§ 100.111(c)(2)(xi)	ATTEST	Propulsion present — requires reliability analysis
FREQ-01	Comply with all applicable technical/operational rules	§ 100.112(c)(1)	ATTEST	Requires operator attestation
FREQ-02	Operate under ITU coordination procedures	§ 100.112(c)(2)	ATTEST	Requires operator attestation
FREQ-03	Spacecraft can be commanded to cease transmissions	§ 100.112(c)(3)	ATTEST	Requires operator attestation

4. Targeted Review Category Pre-Screen (§ 100.136(b))

Targeted Category	FCC Rule	Triggered	Action Required
Failure to Certify	§ 100.136(b)(1)	CLEAR	No failed certifications. <i>None</i>
Waiver Request	§ 100.136(b)(2)	CLEAR	No waivers requested. <i>None</i>
Market Access	§ 100.136(b)(3)	CLEAR	U.S.-licensed operator. <i>None</i>
Foreign Ownership	§ 100.136(b)(4)	CLEAR	Foreign ownership: 0.0%. <i>None</i>
Processing Round	§ 100.136(b)(5)	YES	Applicant opts into NGSO processing round. <i>§ 100.141 processing round procedures</i>
Spectral Constraints	§ 100.136(b)(6)	CLEAR	Requires manual spectrum analysis — not auto-detectable in MVP. <i>Manual review recommended</i>
Federal Coordination	§ 100.136(b)(7)	CLEAR	No federal band allocations requested. <i>None</i>

5. Strategic Counsel Filing Recommendation

Regulatory Counsel Assessment:

RECOMMENDATION: Generally favorable for Part 100 filing. 1 targeted review category(ies) triggered, which will require focused review but should not prevent grant. Consider filing under Part 100 once effective. BOND NOTE: Processing round participation requires \$10M surety bond (de-escalating with deployment). Evaluate financial impact.

Legal & Regulatory Compliance Notice

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